Lili Ju

Curriculum Vitae

Department of Mathematics
University of South Carolina
1523 Greene Street
Columbia, SC 29208, USA

Voice: 803-576-5797
Fax: 803-777-3783
Email: ju@math.sc.edu
http://www.math.sc.edu/~ju

Education

Ph.D. in *Applied Mathematics*, 2002/5, Iowa State University
M.S. in *Computational Mathematics*, 1998/7, Chinese Academy of Sciences
B.S. in *Mathematics*, 1995/7, Wuhan University, China

Working Experiences

2013/1 – Present Professor
Department of Mathematics, University of South Carolina

2008/8 – 2012/12 Associate Professor
Department of Mathematics, University of South Carolina

2004/8 – 2008/8 Assistant Professor
Department of Mathematics, University of South Carolina

2002/9 – 2004/8 Industrial Postdoctoral Fellow
Institute for Mathematics and its Applications, University of Minnesota

1998/8 – 2002/8 Graduate Teaching/Research Assistant
Department of Mathematics and Computing Center, Iowa State University

Awards and Honors

- *Plenary Speaker*, Twelfth China National Conference on Computational Mathematics, 2019.
- *ACM Gordon Bell Prize finalist* in High Performance Computing, 2016.
- *USC Featured Scholar of June 2012*, University of South Carolina, 2012.
- *J.J.L. Hinrichsen Award* for outstanding research in Applied Math., Iowa State Univ., 2002.
- *Research Excellence Award* for graduate students, Iowa State University, 2002.

Professional Activities

- Member, Society for Industrial and Applied Mathematics (SIAM)
- Member, American Mathematics Society (AMS)
- President (2008-2009), Secretary-Treasurer (2007-2008), SIAM Southeastern Atlantic Section

Editorship

- Associate Editor of *SIAM Journal on Numerical Analysis*, 2012/1 – 2017/12.
- Associate Editor of *Numerical Mathematics: Theory, Methods and Applications*, 2021/1 – Present.
- Editorial board of *ISRN Applied Mathematics*, 2011/1 – 2012/12.
- Guest Editor of *Numerical Mathematics: Theory, Methods and Applications* (NMTMA) for the special issue (Volume **3**, Number 2, 2010) on "Centroidal Voronoi Tessellations: Theory, Algorithms and Applications".

Research Interests

Numerical algorithms and analysis
Computational geoscience
Computational materials science
Imaging and computer vision
Deep learning and applications
High performance computing

Grant Support

- ☐ Department of Energy, DE-SC0020270, *"Efficient and Scalable Time Stepping Algorithms and Reduced Order Modeling for Ocean System Simulations"*, PI, 9/1/2019-8/31/2022, $375,000.
- ☐ National Science Foundation, DMS-1818438, *"Study on Localized Exponential Time Differencing Methods for Evolution Partial Differential Equations"*, PI, 8/1/2018-7/31/2021, $150,000.
- ☐ National Science Foundation, DMS-1748357, *"The Ninth Annual Graduate Student Mini-conference in Computational Mathematics"*, Co-PI (PI: Zhu Wang), 1/1/2018-12/31/2018, $7,280. (**Conference grant**)
- ☐ Department of Energy, DE-SC0016540, *"Grid Generation, Coupling Strategies, and Spatially-Dependent Time Stepping for Ocean Tidal/Estuary Systems and Other ESM Components"*, PI, 9/1/2016-8/31/2019, $584,495.
- ☐ National Science Foundation, DMS-1521965, *"Fast and Stable Compact Exponential Time Difference Based Methods for Some Parabolic Equations"*, PI, 8/1/2015-7/31/2018, $200,954.
- ☐ National Science Foundation, DMS-1215659, *"Numerical Improvements, Mesh Adaptation and Parameter Identification for Parallel Finite Element Stokes Ice Sheet Modeling"*, PI, 8/1/2012-7/31/2015, $157,618.
- ☐ Department of Energy, DE-SC0008087-ER65393, *"Predicting Ice Sheet and Climate Evolution at Extreme Scales (PISCEES)"*, PI, 6/15/2012-6/14/2017, $348,420.
- ☐ National Science Foundation, DMS-0913491, *"Study on Algorithms and Applications of Centroidal Voronoi Tessellations"*, PI, 9/1/2009-8/31/2012, $180,000.
- ☐ Society for Industrial and Applied Mathematics, *"The 33rd SIAM Southeastern-Atlantic Section Annual Meeting"*, PIs (L. Szekely and L. Ju), 2009, $4,180. (**Conference grant**)
- ☐ Department of Energy, DE-FG02-07ER64431, *"New Grid and Discretization Technologies for Ocean and Ice Simulations"*, PI, 7/15/2007-7/14/2011, $236,871.
- ☐ National Science Foundation, DMS-0609575, *"Some Problems on Analyses and Applications of Centroidal Voronoi Tessellations"*, PI, 8/1/2006-7/31/2009, $122,781.
- ☐ South Carolina Research Foundation, RPS-13060-06-12306, *"Some Applications of Centroidal Voronoi Tessellations"*, PI, 4/1/2006-6/30/2007, $5,000.

Publications

Dissertation and Thesis

- *"Probabilistic and Parallel Algorithms for Centroidal Voronoi Tessellations with Application to Meshless Computing and Numerical Analysis on Surface"*, **Ph.D. Dissertation**, Iowa State University, 2002.
- *"Parallel Implementation of a Pressure-Correction Projection Method for the Unsteady Incompressible Navier-Stokes Equations"*, **M.S. Thesis**, Institute of Computational Mathematics and Scientific/Engineering Computing, Chinese Academy of Sciences, 1998.

Research Papers

Journal articles

1. L. Ju and L.-B. Zhang, Parallel implementation of a pressure-correction projection method for the unsteady incompressible Navier-Stokes equations, *Mathematica Numerica Sinica*, Vol. **20**, pp. 325-336, 1998.
2. Q. Du, M. Gunzburger and L. Ju, Meshfree, probabilistic determination of point sets and support regions for meshless computing, *Computer Methods in Applied Mechanics and Engineering*, Vol. **191**, pp. 1349-1366, 2002.
3. L. Ju, Q. Du and M. Gunzburger, Probabilistic methods for centroidal Voronoi tessellations and their parallel implementations, *Parallel Computing*, Vol. **28**, pp. 1477-1500, 2002.
4. Q. Du, M. Gunzburger and L. Ju, Constrained centroidal Voronoi tessellations for surfaces, *SIAM Journal on Scientific Computing*, Vol. **24**, pp. 1488-1506, 2003.
5. G. Luecke, M. Kraeva and L. Ju, Comparing the performance of MPICH with Cray's MPI and with SGI's MPI, *Concurrency and Computation: Practice and Experience*, Vol. **15**, pp. 779-802, 2003.
6. Q. Du, M. Gunzburger and L. Ju, Voronoi-based finite volume methods, optimal Voronoi meshes, and PDEs on the sphere, *Computer Methods in Applied Mechanics and Engineering*, Vol. **192**, pp. 3933-3957, 2003.
7. M. Gunzburger, L.S. Hou and L. Ju, Approximation of exact boundary controllability problems for the 1-D wave equation by optimization-based methods, *Contemporary Mathematics*, Vol. **330**, pp. 133-153, 2003.
8. Q. Du and L. Ju, Numerical simulation of the quantized vortices in a thin superconducting hollow sphere, *Journal of Computational Physics*, Vol. **201**, pp. 511-530, 2004.
9. Q. Du and L. Ju, Approximations of a Ginzburg-Landau model for superconducting hollow spheres based on spherical centroidal Voronoi tessellations, *Mathematics of Computation*, Vol. **74**, pp. 1257-1280, 2005.
10. Q. Du and L. Ju, Finite volume methods on spheres and spherical centroidal Voronoi meshes, *SIAM Journal on Numerical Analysis*, Vol. **43**, pp. 1673-1692, 2005.
11. L. Ju, M. Hurdal, K. Rehm, K. Schaper, J. Stern and D. Rottenberg, Quantitative evaluation of three cortical surface flattening methods, *Neuroimage*, Vol. **28**, pp. 869-880, 2005.
12. Q. Du, M. Emelianenko and L. Ju, Convergence of the Lloyd algorithm for computing centroidal Voronoi tessellations, *SIAM Journal on Numerical Analysis*, Vol. **44**, pp. 102-119, 2006.
13. Q. Du, M. Gunzburger, L. Ju and X. Wang, Centroidal Voronoi tessellation algorithms for image compression, segmentation, and multichannel restoration, *Journal of Mathematical Imaging and Vision*, Vol. **24**, pp. 177-194, 2006.
14. M. Gunzburger, L.S. Hou and L. Ju, A numerical method for exact boundary controllability problems for the 2-D wave equation, *Computers & Mathematics with Applications*, Vol. **51**, pp. 721-750, 2006.
15. L. Ju, M. Gunzburger and W. Zhao, Adaptive finite element methods for elliptic PDEs based on conforming centroidal Voronoi Delaunay triangulations, *SIAM Journal on Scientific Computing*, Vol. **28**, pp. 2023-2053, 2006.
16. L. Ju, Conforming centroidal Voronoi Delaunay triangulation for quality mesh generation, *International Journal of Numerical Analysis and Modeling*, Vol. **4**, pp. 531-547, 2007.
17. M. Emelianenko, L. Ju and A. Rand, Nondegeneracy and weak global convergence of the Lloyd algorithm in $\mathbb{R}^d$, *SIAM Journal on Numerical Analysis*, Vol. **46**, pp. 1423-1441, 2008.
18. W. Zhao, L. Tian and L. Ju, Convergence analysis of a splitting scheme for stochastic differential equations, *International Journal of Numerical Analysis and Modeling*, Vol. **5**, pp. 673-692, 2008.

19. L. Ju, H.-C. Lee and L. Tian, Numerical simulations of the steady Navier-Stokes equations using adaptive meshing schemes, *International Journal for Numerical Methods in Fluids*, Vol. **56**, pp. 703-721, 2008.
20. T. Ringler, L. Ju and M. Gunzburger, A multi-resolution method for climate system modeling, *Ocean Dynamics*, Vol. **58**, pp. 475-498, 2008.
21. H. Nguyen, J. Burkardt, M. Gunzburger, L. Ju and Y. Saka, Constrained CVT meshes and a comparison of triangular mesh generators, *Computational Geometry: Theory and Applications*, Vol. **42**, pp. 1-19, 2009.
22. L. Ju and Q. Du, A finite volume method on general surfaces and its error estimates, *Journal of Mathematical Analysis and Applications*, Vol. **352**, pp. 645-668, 2009.
23. Q. Du, L. Ju and L. Tian, Analysis of a mixed finite-volume discretization for fourth-order equations on general surfaces, *IMA Journal of Numerical Analysis*, Vol. **29**, pp. 376-403, 2009.
24. L. Ju ,W. Wu and W. Zhao, Adaptive finite volume methods for steady convection-diffusion equations with mesh optimization, *Discrete and Continuous Dynamical Systems B*, Vol. **11**, pp. 669-690, 2009.
25. H. Nguyen, M. Gunzburger, L. Ju and J. Burkardt, Adaptive anisotropic meshing for steady convection-dominated problems, *Computer Methods in Applied Mechanics and Engineering*, Vol. **198**, pp. 2964-2981, 2009.
26. J. Wang, L. Ju and X. Wang, An edge-weighted centroidal Voronoi tessellation model for image segmentation, *IEEE Transactions on Image Processing*, Vol. **18**, pp. 1844-1858, 2009.
27. L. Ju, L. Tian and D. Wang, A posteriori error estimates for finite volume approximations of elliptic equations on general surfaces, *Computer Methods in Applied Mechanics and Engineering*, Vol. **198**, pp. 716-726, 2009.
28. L. Zhu, Y. Wang, L. Ju and D. Wang, A variational phase field method for curve smoothing, *Journal of Computational Physics*, Vol. **229**, pp. 2390-2400, 2010.
29. Q. Du, M. Gunzburger and L. Ju, Advances in studies and applications of centroidal Voronoi tessellations, *Numerical Mathematics: Theory, Methods and Applications*, Vol. **3**, pp. 119-142, 2010.
30. E. Choate, M. Forest and L. Ju, Effects of strong anchoring on the dynamic moduli of heterogeneous nematic polymers II: Oblique anchoring angles, *Rheologica Acta*, Vol. **49**, pp. 335-347, 2010.
31. W. Zhao, G. Zhang and L. Ju, A stable multi-step scheme for solving backward stochastic differential equations, *SIAM Journal on Numerical Analysis*, Vol. **48**, pp. 1369-1394, 2010.
32. H. Zhang, L. Ju, M. Gunzburger, T. Ringler and S. Price, Coupled models and parallel simulations for three dimensional full-Stokes ice sheet modeling, *Numerical Mathematics: Theory, Methods and Applications*, Vol. **4**, pp. 359-381, 2011.
33. Q. Du, L. Ju and L. Tian, Finite element approximation of the Cahn-Hilliard equations on surfaces, *Computer Methods in Applied Mechanics and Engineering*, Vol. **200**, pp. 2458-2470, 2011.
34. J. Wang, L. Ju and X. Wang, Image segmentation using local variation and edge-weighted centroidal Voronoi tessellations, *IEEE Transactions on Image Processing*, Vol. **20**, pp. 3242-3256, 2011.
35. T. Ringler, D. Jacobsen, M. Gunzburger, L. Ju, M. Duda and W. Skamarock, Exploring a multi-resolution modeling approach within the shallow-water equations, *Monthly Weather Review*, Vol. **139**, pp. 3348-3368, 2011.
36. W. Leng, L. Ju, M. Gunzburger, S. Price and T. Ringler, A parallel high-order accurate finite element nonlinear Stokes ice sheet model and benchmark experiments, *Journal of Geophysical Research*, Vol. **117**, F01001 (24 pages), 2012.

37. L. Ju, L. Tian, X. Xiao and W. Zhao, Covolume-upwind finite volume approximations for linear elliptic partial differential equations, *Journal of Computational Physics*, Vol. **231**, pp. 6097-6120, 2012.

38. Y. Wang, L. Ju, D. Wang and X. Wang, Generalized edge-weighted centroidal Voronoi tessellations for geometry processing, *Computers & Mathematics with Applications*, Vol. **64**, pp. 2663-2681, 2012.

39. W. Leng, L. Ju, M. Gunzburger and S. Price, Manufactured solutions and the verification of three-dimensional Stokes ice-sheet models, *The Cryosphere*, Vol. **7**, pp. 19-29, 2013.

40. W. Zhao, Y. Li and L. Ju, Error estimates of the Crank-Nicolson scheme for solving backward stochastic differential equations, *International Journal of Numerical Analysis and Modeling*, Vol. **10**, pp. 876-898, 2013.

41. Q. Du, L. Ju, L. Tian and K. Zhou, A posteriori error analysis of finite element method for linear nonlocal diffusion and peridynamic models, *Mathematics of Computation*, Vol. **82**, pp. 1889-1922, 2013.

42. Y. Cao, L. Ju, Y. Zhou and S. Wang, 3D superalloy grain segmentation using a multichannel edge-weighted centroidal Voronoi tessellation algorithm, *IEEE Transactions on Image Processing*, Vol. **22**, pp. 4123-4135, 2013.

43. W. Zhao, W. Zhang and L. Ju, A numerical method and its error estimates for the decoupled forward-backward stochastic differential equations, *Communications in Computational Physics*, Vol. **15**, pp. 618-646, 2014.

44. W. Leng, L. Ju, Y. Xie, T. Cui and M. Gunzburger, Finite element three-dimensional Stokes ice sheet dynamics model with enhanced local mass conservation, *Journal of Computational Physics*, Vol. **274**, pp. 299-311, 2014.

45. L. Ju, X. Liu, and W. Leng, Compact implicit integration factor methods for a family of semilinear fourth-order parabolic equations, *Discrete and Continuous Dynamical Systems B*, Vol. **19**, pp. 1667-1687, 2014.

46. W. Leng, L. Ju, M. Gunzburger and S. Price, A parallel computational model for three-dimensional, thermo-mechanical Stokes flow simulations of glaciers and ice sheets, *Communications in Computational Physics*, Vol. **16**, pp. 1056-1080, 2014.

47. L. Ju, J. Zhang, L. Zhu and Q. Du, Fast explicit integration factor methods for semilinear parabolic equations, *Journal of Scientific Computing*, Vol. **62**, pp. 431-455, 2015.

48. H. Tian, L. Ju and Q. Du, Nonlocal convection-diffusion problems and finite element approximations, *Computer Methods in Applied Mechanics and Engineering*, Vol. **289**, pp. 60-78, 2015.

49. Y. Zhou, L. Ju and S. Wang, Multiscale superpixels and supervoxels based on hierarchical edge-Weighted centroidal Voronoi tessellation, *IEEE Transactions on Image Processing*, Vol. **24**, pp. 3834-3845, 2015.

50. T. Zhang, L. Ju, W. Leng, S. Price and M. Gunzburger, Thermomechanically coupled glacier modeling for land-terminating glaciers: A comparison of two-dimensional, first-order and three-dimensional, full Stokes approaches, *Journal of Glaciology*, Vol. **61**, pp. 702-712, 2015.

51. L. Ju, J. Zhang and Q. Du, Fast and accurate algorithms for simulating coarsening dynamics of Cahn-Hilliard equations, *Computational Materials Science*, Vol. **108**, pp. 272-282, 2015.

52. W. Zhao, W. Zhang and L. Ju, A multistep scheme for decoupled forward-backward stochastic differential equations, *Numerical Mathematics: Theory, Methods and Applications*, Vol. **9**, pp. 262-288, 2016.

53. X. Zhang, M. Gunzburger and L. Ju, Nodal-type collocation method for hypersingular integral equations and nonlocal diffusion problems, *Computer Methods in Applied Mechanics and Engineering*, Vol. **299**, pp. 401-420, 2016.

54. X. Zhang, M. Gunzburger and L. Ju, Quadrature rules for finite element approximations of 1D nonlocal problems, *Journal of Computational Physics*, Vol. **310**, pp. 213-236, 2016.
55. X. Wang, L. Ju and Q. Du, Efficient and stable exponential time differencing Runge-Kutta methods for phase field elastic bending energy models, *Journal of Computational Physics*, Vol. **316**, pp. 21-38, 2016.
56. L. Zhu, L. Ju and W. Zhao, Fast high-order compact exponential time differencing Runge-Kutta methods for second-order semilinear parabolic equations, *Journal of Scientific Computing*, Vol. **67**, pp. 1043-1065, 2016.
57. X. Fan, L. Ju, X. Wang and S. Wang, A fuzzy edge-weighted centroidal Voronoi tessellation model for image segmentation, *Computers & Mathematics with Applications*, Vol. **71**, pp. 2272-2284, 2016.
58. X. Yang and L. Ju, Efficient linear schemes with unconditionally energy stability for the phase field elastic bending energy model, *Computer Methods in Applied Mechanics and Engineering*, Vol. **315**, pp. 691-712, 2017.
59. T. Zhang, S. Price, L. Ju, W. Leng, J. Brondex, G. Durand and O. Gagliardini, A comparison of two Stokes ice sheet models applied to the Marine Ice Sheet Model Intercomparison Project for plan view models (MISMIP3d), *The Cryosphere*, Vol. **11**, pp. 179-190, 2017.
60. X. Yang and L. Ju, Linear and unconditionally energy stable schemes for the binary fluid-surfactant phase field model, *Computer Methods in Applied Mechanics and Engineering*, Vol. **318**, pp. 1005-1029, 2017.
61. H. Tian, L. Ju and Q. Du, A conservative nonlocal convection-diffusion model and asymptotically compatible finite difference discretization, *Computer Methods in Applied Mechanics and Engineering*, Vol. **320**, pp. 46-67, 2017.
62. L. Ju and Z. Wang, Exponential time differencing gauge method for incompressible viscous flows, *Communications in Computational Physics*, Vol. **22**, pp. 517-541, 2017.
63. L. Ju, X. Li, Z. Qiao and H. Zhang, Energy stability and error estimates of exponential time differencing schemes for the epitaxial growth model without slope selection, *Mathematics of Computation*, Vol. **87**, pp. 1859-1885, 2018.
64. Q. Du, L. Ju, X. Li and Z. Qiao, Stabilized linear semi-implicit schemes for the nonlocal Cahn-Hilliard equation, *Journal of Computational Physics*, Vol. **363**, pp. 39-54, 2018.
65. S. Duo, L. Ju and Y. Zhang, A fast algorithm for solving the space-time fractional diffusion equation, *Computers & Mathematics with Applications*, Vol. **75**, pp. 1929-1941, 2018.
66. H. Li, L. Ju, C. Zhang and Q. Peng, Unconditionally energy stable linear schemes for a diffuse interface model with Peng-Robinson equation of state, *Journal of Scientific Computing*, Vol. **75**, pp. 993-1015, 2018.
67. S. Li, L. Luo, Z.J. Wang and L. Ju, An exponential time-integrator scheme for steady and unsteady inviscid flows, *Journal of Computational Physics*, Vol. **365**, pp. 206-225, 2018.
68. X. Zhang, J. Wu and L. Ju, An accurate and asymptotically compatible collocation method for nonlocal diffusion problems, *Applied Numerical Mathematics*, Vol. **133**, pp. 52-68, 2018.
69. Q. Chen and L. Ju, Conservative finite volume schemes for solving the quasi-geostrophic equation on coastal-conforming unstructured primal-dual meshes, *Quarterly Journal of the Royal Meteorological Society*, Vol. **144**, pp. 1106-1122, 2018.
70. H. Feng, Y. Gao, L. Ju, H. Tian and X. Zhang, Nodal-type Newton-Cotes rules for fractional order hypersingular integrals, *East Asian Journal of Applied Mathematics*,Vol. **8**, pp. 697-714, 2018.
71. H. Yang, M. Gunzburger and L. Ju, Fast spherical centroidal Voronoi mesh generation: a Lloyd-preconditioned LBFGS method in parallel, *Journal of Computational Physics*, Vol. **367**, pp. 235-252, 2018.

72. T.-T.-P. Hoang, L. Ju and Z. Wang, Overlapping localized exponential time differencing methods for diffusion problems, *Communications in Mathematical Sciences*, Vol. **16**, pp. 1531-1555, 2018.

73. H. Feng, Y. Gao, L. Ju and X. Zhang, A new collocation method for solving certain Hadamard finite-part integral equations, *International Journal of Numerical Analysis and Modeling*, Vol. **16**, pp. 240-254, 2019.

74. Q. Du, L. Ju and J. Lu, A discontinuous Galerkin method for one-dimensional time-dependent nonlocal diffusion problems, *Mathematics of Computation*, Vol. **88**, pp. 123-147, 2019.

75. T.-T.-P. Hoang, W. Leng, L. Ju, Z. Wang and K. Pieper, Conservative explicit local time-stepping schemes for the shallow water equations, *Journal of Computational Physics*, Vol. **382**, pp. 152-176, 2019.

76. Q. Du, L. Ju and J. Lu, Analysis of fully discrete approximations for dissipative systems and application to time-dependent nonlocal diffusion problems, *Journal of Scientific Computing*, Vol. **78**, pp. 1438-1466, 2019.

77. J. Huang, L. Ju and B. Wu, A fast compact exponential time differencing method for semi-linear parabolic equations with Neumann boundary conditions, *Applied Mathematics Letters*, Vol. **94**, pp. 257-265, 2019.

78. Q. Du, L. Ju, X. Li and Z. Qiao, Maximum principle preserving exponential time differencing schemes for the nonlocal Allen-Cahn equation, *SIAM Journal on Numerical Analysis*, Vol. **57**, pp. 875-898, 2019.

79. C. Zhang, H. Li, X. Zhang and L. Ju, Linear and unconditionally energy stable schemes for the multi-component two-phase diffuse interface model with Peng-Robinson equation of state, *Communications in Computational Physics*, Vol. **26**, pp. 1071-1097, 2019.

80. W. Leng and L. Ju, An additive overlapping domain decomposition method for the Helmholtz equation, *SIAM Journal on Scientific Computing*, Vol. **41**, pp. A1252-A1277, 2019.

81. H. Gao, L. Ju and W. Xie, A stabilized semi-implicit Euler gauge-invariant method for the time-dependent Ginzburg-Landau equations, *Journal of Scientific Computing*, Vol. **80**, pp. 1083-1115, 2019.

82. X. Li, L. Ju and X.-C. Meng, Convergence analysis of exponential time differencing schemes for the Cahn-Hilliard equation, *Communications in Computational Physics*, Vol. **26**, pp. 1510-1529, 2019.

83. J. Huang, L. Ju and B. Wu, A fast compact time integrator method for a family of general order semilinear evolution equations, *Journal of Computational Physics*, Vol. **393**, pp. 313-336, 2019.

84. H. Gao, L. Ju, R. Duddu and H. Li, An efficient second-order linear scheme for the phase field model of corrosive dissolution, *Journal of Computational and Applied Mathematics*, Vol. **367**, Article #112472 (16 pages), 2020.

85. H. Tian, J. Zhang and L. Ju, A spectral collocation method for nonlocal diffusion equations with volume constrained boundary conditions, *Applied Mathematics and Computation*, Vol. **370**, Article #124930 (15 pages), 2020.

86. Q. Du, L. Ju, J. Lu and X. Tian, A discontinuous Galerkin method with penalty for one-dimensional nonlocal diffusion problems, *Communications on Applied Mathematics and Computation*, Vol. **2**, pp. 31-55, 2020.

87. H. Gao, L. Ju, X. Li and R. Duddu, A space-time adaptive finite element method with exponential time integrator for the phase field model of pitting corrosion, *Journal of Computational Physics*, Vol. **406**, Article #109191 (20 pages), 2020.

88. T.-T.-P. Hoang, L. Ju and Z. Wang, Nonoverlapping localized exponential time differencing methods for diffusion problems, *Journal of Scientific Computing*, Vol. **82**, Article #37, 2020.

89. T.-T.-P. Hoang, L. Ju, W. Leng and Z. Wang, High order explicit local time-stepping methods for hyperbolic conservation laws, *Mathematics of Computation*, Vol. **89**, pp. 1807-1842, 2020.

90. L. Ju, W. Leng, Z. Wang and S. Yuan, Numerical investigation of ensemble methods with block iterative solvers for evolution problems, *Discrete and Continuous Dynamical Systems B*, Vol. **25**, pp. 4905-4923, 2020.

91. X. Meng, T.-T.-P. Hoang, Z. Wang and L. Ju, Localized exponential time differencing methods for shallow water equations: algorithms and numerical study, *Communications on Computational Physics*, Vol. **29**, pp. 80-110, 2021.

92. W. Leng and L. Ju, A diagonal sweeping domain decomposition method with source transfer for the Helmholtz equation, *Communications on Computational Physics*, Vol. **29**, pp. 357-398, 2021.

93. X. Li, L. Ju and T.-T.-P. Hoang, Domain decomposition-based exponential time differencing methods for semilinear parabolic equations, *BIT Numerical Mathematics*, to appear. DOI: 10.1007/s10543-020-00817-0

94. Q. Du, L. Ju, X. Li and Z. Qiao, Maximum bound principles for a class of semilinear parabolic equations and exponential time differencing schemes, *SIAM Review*, to appear. DOI: 10.1137/19M1243750

95. J. Li, X. Li, L. Ju and X. Feng, Stabilized integrating factor Runge-Kutta method and unconditional preservation of maximum bound principle, *SIAM Journal on Scientific Computing*, to appear.

96. Q. Chen, L. Ju and R. Temam, Conservative numerical schemes with optimal dispersive wave relations - Part I. Derivations and analyses, *Numerische Mathematik*, to appear.

97. J. Li, L. Ju, Y. Cai and X. Feng, Unconditionally maximum principle preserving linear schemes for the conservative Allen-Cahn equation with nonlocal constraint, *Journal of Scientific Computing*, to appear.

98. L. Ju, X. Li, Z. Qiao and J. Yang, Maximum bound principle preserving integrating factor Runge-Kutta methods for semilinear parabolic equations, *submitted*.

99. W. Leng and L. Ju, Trace transfer-based diagonal sweeping domain decomposition method for the Helmholtz equation: algorithms and convergence analysis, *submitted*.

100. K. Jiang, L. Ju, J. Li and X. Li, Unconditionally stable exponential time differencing schemes for the mass-conserving Allen-Cahn equation with nonlocal and local effects, *submitted*.

101. Y. Wang, K. Luo, Z. Chen, L. Ju and T. Guan, DeepFusion: A simple way to improve traditional multi-view stereo methods using deep learning, *submitted*.

102. H. Tian, M. Bi and L. Ju, A dual-horizon nonlocal diffusion model and its finite element discretization, *submitted*.

103. Q. Chen, L. Ju and R. Temam, Conservative numerical schemes with optimal dispersive wave relations - Part II. Numerical evaluations, *submitted*.

104. X. Zhang, J. Wang, Y. Zhu, L. Ju, J. Yang and Y. Qian, GW-PINN: A deep learning method for solving regional groundwater problems, *submitted*.

105. R. Lan, W. Leng, Z. Wang, L. Ju, and M. Gunzburger, Parallel exponential time differencing methods for geophysical flow simulations, *submitted*.

Conference proceedings

106. M. Gunzburger, L.S. Hou and L. Ju, A numerical method for controllability problems for the wave equation, in Hyperbolic Problems: Theory, Numerics, Applications, ed. by E. Tadmor and T. Hou, *Proceedings of Ninth International Conference on Hyperbolic Problems*, pp. 557-568, Pasadena, CA, March, 2002.

107. L. Ju, J. Stern, K. Rehm, K. Schaper, M. Hurdal and D. Rottenberg, Cortical surface flattening using least square conformal mapping with minimal metric distortion, *Proceedings of Second IEEE International Symposium on Biomedical Imaging* (ISBI'2004), pp. 77-80, Arlington, VA, April, 2004.
108. E. Choate, M. Gorrest, Z. Cui and L. Ju, Effects of director angle anchoring conditions on the dynamic moduli of heterogeneous nematic polymers, *Proceedings of XVth International Congress on Rheology*, pp. 481-483, Monterey, CA, August, 2008.
109. P. Dalal, L. Ju, M. McLaughlin, X. Zhou, H. Fujita and S. Wang, 3D open-surface shape correspondence for statistical shape modeling: Identifying topologically consistent landmarks, *Proceedings of Twelfth IEEE International Conference on Computer Vision* (ICCV'2009), pp. 1857-1864, Kyoto, Japan, September-October, 2009.
110. Y. Cao, L. Ju, Q. Zou, C.-Z. Qu and S. Wang, A multichannel edge-weighted centroidal Voronoi tessellation algorithm for 3D super-alloy image segmentation, *Proceedings of Twenty-Fourth IEEE Conference on Computer Vision and Pattern Recognition* (CVPR'2011), pp. 17-24, Colorado Springs, CO, June, 2011.
111. Y. Cao, L. Ju and S. Wang, Grain segmentation in 3D superalloy images using multichannel EWCVT under human annotation constraints, *Proceedings of Twelfth European Conference on Computer Vision* (ECCV'2012), Volume **3**, pp. 244-257, Firenze, Italy, October, 2012.
112. Y. Zhou, L. Ju, Y. Cao, J. Waggoner, Y. Lin, J. Simmon and S. Wang, Edge-weighted centroid Voronoi tessellation with propagation of consistency constraint for 3d grain segmentation in microscopic superalloy images, *Proceedings of Tenth IEEE Workshop on Perception Beyond the Visible Spectrum* (PBVS'2014), pp. 258-265, Columbus, OH, June, 2014.
113. Y. Zhou, L. Ju and S. Wang, Multiscale superpixels and supervoxels based on hierarchical edge-Weighted centroidal Voronoi tessellation, *Proceedings of IEEE Winter Conference on Applications of Computer Vision* (WACV'2015), pp. 1076-1083, Waikoloa Beach, HI, January, 2015.
114. J. Zhang, C. Zhou, Y. Wang, L. Ju, Q. Du, X. Chi, D. Xu, D. Chen, Y. Liu and Z. Liu, Extreme-scale phase field simulations of coarsening dynamics on the Sunway TaihuLight supercomputer, *Proceedings of International Conference for High Performance Computing, Networking, Storage and Analysis* (SC'2016), Article #4 (12 pages), Salt Lake City, UT, USA, November, 2016. (**2016 ACM Gordon Bell Prize finalist**)
115. S. Li, Z.J. Wang, L. Ju and L. Luo, Explicit large time stepping with a second-order exponential time integrator scheme for unsteady and steady flows, *Proceedings of AIAA SciTech Forum and Exposition* (SciTech'2017), Article #0753 (18 pages), Grapevine, Texas, January, 2017.
116. S. Li, Z.J. Wang, L. Ju and L. Luo, Fast time integration of Navier-Stokes equations with an exponential-integrator scheme, *Proceedings of AIAA SciTech Forum and Exposition* (SciTech'2018), Article #0369 (12 pages), Kissimmee, FL, January, 2018.
117. S. Li and L. Ju, Exponential time-marching method for the unsteady Navier-Stokes equations, *Proceedings of AIAA SciTech Forum and Exposition* (SciTech'2019), Article #0907 (11 pages), San Diego, CA, January, 2019.
118. Z. Wu, X. Wu, X. Zhang, S. Wang and L. Ju, Semantic stereo matching with pyramid cost volumes, *Proceedings of Seventeenth International Conference on Computer Vision* (ICCV'2019), pp. 7484-7493, Seoul, Korea, October-November, 2019.
119. Z. Wu, X. Wu, X. Zhang, S. Wang and L. Ju, Spatial correspondence with generative adversarial network: learning depth from monocular videos, *Proceedings of Seventeenth International Conference on Computer Vision* (ICCV'2019), pp. 7494-7504, Seoul, Korea, October-November, 2019.
120. K. Luo, T. Guan, L. Ju, H. Huang and Y. Luo, P-MVSNet: Learning patch-wise matching confidence aggregation for multi-view stereo, *Proceedings of Seventeenth International Conference on Computer Vision* (ICCV'2019), pp. 10452-10461, Seoul, Korea, October-November, 2019.

121. X. Wu, Z. Wu, Y. Zhang, L. Ju and S. Wang, Multi-video temporal synchronization by matching pose features of shared moving subjects, *Proceedings of ICCV Workshop on Moving Camera* (MCMVS'2019), Seoul, Korea, October, 2019.
122. S. Li, L. Ju and H. Si, Adaptive exponential time-marching method for the compressible Navier-Stokes equations, *Proceedings of AIAA SciTech Forum and Exposition* (SciTech'2020), Article # 2033, Orlando, FL, January, 2020.
123. X. Wu, Z. Wu, J. Zhang, L. Ju and S. Wang, SalBSC: A video saliency prediction model based on shuffled attentions and correlation-based ConvLSTM, *Proceedings of Thirty-Fourth AAAI Conference on Artificial Intelligence* (AAAI'2020), pp. 12410-12417, New York, February, 2020.
124. K. Luo, T. Guan, L. Ju, Y. Wang, Z. Chen and Y. Luo, Attention-enhanced multi-view stereo, *Proceedings of Thirty-Third IEEE Conference on Computer Vision and Pattern Recognition* (CVPR'2020), pp. 1590-1599, Seattle, WA, June, 2020.
125. Y Wang, T. Guan, Z. Chen, Y. Luo, K. Luo and L. Ju, Mesh-guided multi-view stereo with pyramid architecture, *Proceedings of Thirty-Third IEEE Conference on Computer Vision and Pattern Recognition* (CVPR'2020), pp. 2039-2048, Seattle, WA, June, 2020.
126. Z. Wu, X. Wu, L. Ju and S. Wang, Binaural audio visual localization, *Proceedings of Thirty-Fifth AAAI Conference on Artificial Intelligence* (AAAI'2021), to appear, February, 2021.
127. X. Wu, Z. Wu, H. Guo, L. Ju and S. Wang, DANNet: A one-stage domain adaption network for unsupervised nighttime semantic segmentation, *Proceedings of Thirty-Fourth IEEE Conference on Computer Vision and Pattern Recognition* (CVPR'2021), to appear. (Accepted as **Oral**)
128. Z. Wu, X. Wu, X. Zhang, S. Wang and L. Ju, Learning depth from single image using depth-aware convolution and stereo knowledge, *Proceedings of IEEE International Conference on Multimedia and Expo* (ICME'2021), to appear, Shenzhen, China. (Accepted as **Oral**)
129. X. Zhang, T. Cheng and L. Ju, Implicit form neural network for learning scalar hyperbolic conservation laws, *Proceedings of Second Conference on Mathematical and Scientific Machine Learning* (MSML'2021), to appear.
130. Z. Wu, X. Wu, X. Zhang, S. Wang and L. Ju, Depth-contextual representation for self-supervised monocular depth estimation, *submitted.*

Book chapters

131. L. Ju, T. Ringler and M. Gunzburger*, Voronoi diagrams and their application to climate and global modeling, in* Numerical Techniques for Global Atmospheric Models*, Lecture Notes in Computational Science and Engineering, Vol.* **80***, pp. 313-342, Ed. by P. Lauritzen, C. Jablonowski, M. Taylor, R. Nair, Springer, 2011.*

Seminars and Colloquia

Invited

1. Applied Mathematics Seminar, Department of Mathematics, University of North Carolina at Chapel Hill, 2002/1
2. Colloquium, Department of Mathematics, University of Pittsburgh, 2002/2
3. IMA/MCIM Industrial Problems Seminar, School of Mathematics, University of Minnesota, 2003/2
4. Iowa PDE Seminar, Department of Mathematics, Iowa State University, 2003/4
5. Applied and Computational Mathematics Seminar, Department of Mathematics, Pennsylvania State University, 2003/5

6. CSCAMM Seminar, Center for Scientific Computation and Mathematical Modeling, University of Maryland at College Park, 2003/10
7. CSIT Seminar Series, School of Computational Science & Information Technology, Florida State University, 2003/11
8. Colloquium, Department of Mathematics, University of South Carolina, 2004/1
9. Colloquium, Department of Mathematics and Statistics, Texas Tech University, 2004/2
10. Faculty Forum in Scientific Computing, University of South Carolina, 2005/2
11. PDEs and Numerical Methods Seminar, Department of Mathematics, Pennsylvania State University, 2005/3
12. Analysis and Computational Mathematics Seminar, Department of Mathematical Sciences, Clemson University, 2005/4
13. SIAM Student Chapter Seminar, Department of Mathematics, University of South Carolina, 2005/9
14. CNA Seminar, Department of Mathematical Sciences, Carnegie Mellon University, 2006/2
15. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, 2006/6
16. Colloquium, Research and Development Center for Parallel Software, Institute of Software, Chinese Academy of Sciences, 2006/6
17. SCS Seminar Series, School of Computational Science, Florida State University, 2006/11
18. Computational Mathematics Seminar, Department of Mathematical Sciences, Clemson University, 2007/4
19. BK21 Seminar, Department of Mathematics, Ajou University, Korea, 2007/5
20. Colloquium, School of Mathematics and System Sciences, Shandong University, China, 2007/5
21. Computational Nano Seminar, NanoCenter, University of South Carolina, 2007/11
22. Applied and Computational Mathematics Seminar, Department of Mathematical Sciences, George Mason University, 2008/3
23. Colloquium, Department of Mathematics, Xiangtan University, China, 2008/5
24. Colloquium, Department of Mathematics and Systems Science, National University of Defense Technology, China, 2008/5
25. Computational and Applied Mathematics Seminar, Department of Mathematics, Purdue University, 2008/10
26. Colloquium, Department of Mathematics and Statistics, Auburn University, 2008/12
27. SIAM Student Chapter Seminar, Department of Mathematics and Statistics, Auburn University, 2008/12
28. Seminar, Division of Mathematical Sciences, Nanyang Technological University, Singapore, 2008/12
29. Seminar, State Key Lab of Computer-Aided Design&Computer Graphics, Zhejiang University, China, 2009/7
30. Seminar, School of Mathematics and System Sciences, Shandong University, China, 2009/7
31. Aubrey Lecture Series, Department of Earth and Ocean Sciences, University of South Carolina, 2009/10
32. Applied Mathematics Seminar, Department of Mathematics, Ohio State University, 2010/4.
33. School of Sciences, Linyi Normal University, China, 2010/6
34. Seminar, School of Mathematical Sciences, Beijing Normal University, China, 2010/6

35. Department of Mathematics, Shanghai Jiaotong University, China, 2010/6
36. CCMA Luncheon Seminar, Department of Mathematics, Pennsylvania State University, 2010/11
37. Computational and Applied Mathematics Colloquium, Department of Mathematics, Pennsylvania State University, 2010/11
38. Applied and Computational Mathematics Seminar, School of Mathematics, Georgia Institute of Technology, 2011/4
39. Seminar, School of Mathematics, Shandong University, China, 2011/6
40. Seminar, Laboratory of Computational Geodynamics, Graduate University of Chinese Academy of Sciences, 2011/6
41. Seminar, Center for Computational Science & Engineering, Peking University, China, 2011/6
42. Seminar, School of Mathematics, Shandong University, China, 2011/6
43. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, 2011/6
44. Seminar, School of Mathematics, Shandong University, China, 2011/12
45. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, 2012/6
46. Seminar, School of Mathematics, Capital Normal University, China, 2012/6
47. Seminar, School of Mathematical Sciences, Beihang University, China, 2012/7
48. Seminar, School of Sciences, Shanghai Normal University, China, 2012/7
49. Applied Mathematics Seminar, Department of Mathematics, Michigan State University, 2012/9
50. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, 2013/5
51. Seminar, School of Mathematical Sciences, University of Science and Technology of China, China 2013/5
52. Seminar, School of Mathematical Sciences, Beihang University, China, 2013/6
53. Seminar, Laboratory of Computational Geodynamics, Graduate University of Chinese Academy of Sciences, 2013/6
54. Colloquium, Department of Mathematics and Statistics, Missouri University of Science and Technology, 2013/11
55. ACM Seminar, Beijing International Center for Mathematical Research, Peking University, China, 2013/12
56. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, 2013/12
57. Seminar, School of Mathematical Sciences, Beihang University, China, 2014/6
58. Colloquium, Department of Mathematical Sciences, Shandong Normal University, China, 2014/5
59. Seminar, School of Mathematics, Shandong University, China, 2014/7
60. Seminar, State Key Lab of Computer-Aided Design&Computer Graphics, Zhejiang University, China, 2014/7
61. Seminar, School of Mathematical Sciences, Beijing Normal University, China, 2014/7
62. Seminar, Morningstar Mathematical Center, Chinese Academy of Sciences, 2014/7
63. Colloquium, Department of Mathematics, University of Alabama, 2014/9
64. Colloquium, Department of Mathematics and Statistics, Old Dominion University, 2014/10
65. CAM Seminar, Department of Mathematics, Iowa State University, 2014/10

66. Seminar, Chinese Network Information Center, Chinese Academy of Sciences, 2014/12
67. Seminar, School of Mathematics, Wuhan University, China, 2015/5
68. Seminar, School of Mathematics, Shanghai Normal University, China, 2015/6
69. Seminar, School of Mathematics, Sichuan University, China, 2015/7
70. Colloquium, Department of Applied Mathematics, Illinois Institute of Technology, 2015/9
71. Colloquium, Department of Applied Mathematics, Hong Kong Polytechnic University, 2015/10
72. Colloquium, Department of Mathematics, University of Tennessee at Chattanooga, 2015/11
73. Seminar, Institute of Mathematical Sciences, Renmin University of China, China, 2015/12
74. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, 2015/12
75. Applied Mathematics Seminar, Department of Mathematics, George Washington University, 2016/3
76. Seminar, State Key Laboratory of Cryospheric Sciences, Chinese Academy of Sciences, 2016/5
77. Colloquium, School of Mathematical Sciences, Lanzhou University, China, 2016/5
78. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, 2016/5
79. Colloquium, Department of Applied Mathematics, Hong Kong Polytechnic University, 2016/6
80. Seminar, Department of Mathematical Sciences, Tsinghua University, China, 2016/6
81. Seminar, School of Mathematics, Shandong University, China, 2016/7
82. Seminar, School of Mathematical Sciences, Beihang University, China, 2016/7
83. Seminar, School of Mathematics, Wuhan University, 2016/7
84. Seminar, School of Applied Mathematics and Sciences, Beijing University of Technology, China, 2016/7
85. Seminar, Department of mathematical sciences, Clemson University, 2016/9
86. Colloquium, Department of Computational Mathematics, Science and Engineering, Michigan State University, 2016/10
87. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, 2016/12
88. Seminar, School of Mathematics, Wuhan University, China, 2016/12
89. Colloquium, Oak Ridge National Laboratory, 2017/2.
90. CAM Seminar, Department of Mathematics, University of Tennessee at Knoxville, 2017/2.
91. Appl. Math. Seminar, Department of Mathematics, University of Wisconsin at Madison, 2017/4
92. Colloquium, School of Science, Wuhan University of Science and Technology, China, 2017/5
93. Seminar, School of Mathematics, Wuhan University, China, 2017/5
94. Colloquium, School of Mathematical Sciences, Electric University of Science and Technology of China, 2017/5
95. Seminar, Center for Mathematical Sciences, Huazhong University of Science and Technology, China, 2017/7
96. Seminar, School of Mathematical Sciences, Shanghai Jiao Tong University, China, 2017/7
97. Seminar, Department of Mathematical Sciences, Michigan Technological University, 2017/9
98. Seminar, School of Mathematical Sciences, Jilin University, China, 2017/9

99. Seminar, School of Mathematics, Shandong University, China, 2017/10
100. Colloquium, School of Mathematics, Shandong Normal University, China, 2017/10
101. Seminar, School of Mathematical Sciences, Beihang University, China, 2017/10
102. Seminar, Department of Mathematics, University of Hong Kong, Hong Kong, 2017/12
103. Seminar, School of Mathematics and Statistics, Central China Normal University, China, 2017/12
104. Seminar, Institute of Computational Mathematics & Scientific/Engineering Computing, Chinese Academy of Sciences, 2018/1
105. Seminar, China Network Information Center, Chinese Academy of Sciences, 2018/1
106. Seminar, School of Sciences, Shanghai Normal University, China, 2018/1
107. Seminar, School of Mathematical Sciences, Tongji University, China, 2018/1
108. Seminar, School of Mathematical Sciences, Shanghai Jiao Tong University, China, 2018/4
109. Colloquium, School of Mathematical Sciences, Beihang University, China, 2018/4
110. Seminar, School of Sciences, Nanchang University, China, 2018/4
111. Seminar, School of Mathematical Sciences, Ocean University of China, China, 2018/4
112. Seminar, School of Mathematical Sciences, Beijing Normal University, China, 2018/5
113. Seminar, China Network Information Center, Chinese Academy of Sciences, 2018/5
114. Seminar, Department of Mathematics, Southern University of Science and Technology of China, 2018/6
115. Seminar, School of Data and Computer Science, Sun Yat-sen University , China, 2018/6
116. Seminar, School of Applied Mathematics and Sciences, Beijing University of Technology, China, 2018/7
117. Applied Mathematics Seminar, Department of Mathematics, Ohio State University, 2018/10
118. Seminar, School of Mathematics, Shandong University, China, 2018/10
119. Colloquium, School of Mathematics and Statistics, Huazhong University of Science and Technology, China, 2019/1
120. Seminar, School of Mathematics and Statistics, Xuchang University, China, 2019/1
121. CAM Seminar, Department of Mathematics and Statistics, Mississippi State University, 2019/4
122. Seminar, Yau Mathematical Sciences Center, Tsinghua University, China, 2019/5
123. Colloquium, School of Mathematics and Statistics, Zhengzhou University, China, 2019/5
124. Seminar, School of Mathematics and Econometrics, Hunan University, China, 2019/5
125. Seminar, School of Mathematics, Shandong University, China, 2019/5
126. Seminar, School of Mathematics and Statistics, Linyi University, China, 2019/5
127. Seminar, School of Automation and Electrical Engineering, Linyi University, China, 2019/5
128. Seminar, School of Mathematical Sciences, Ocean University of China, China, 2019/5
129. Seminar, School of Sciences, Shanghai Normal University, China, 2019/7
130. Seminar, School of Mathematical Sciences, East China Normal University, China, 2019/7
131. Seminar, School of Mathematical Sciences, Shanghai Jiao Tong University, China, 2019/7
132. Colloquium, Department of Mathematics and Statistics, Texas Tech University, 2019/10
133. Seminar, Department of Mathematics, Ajou University, Korea, 2019/11
134. Seminar, School of Mathematical Sciences, Beihang University, China, 2019/12

135. Colloquium, School of Computer Science and Engineering, South China University of Technology, China, 2019/12
136. Seminar, Research Center for Applied Mathematics and Interdisciplinary Studies, South China Normal University, China, 2019/12
137. Seminar, Department of Applied Mathematical, Hong Kong Polytechnic University, Hong Kong, 2020/5 (Online talk)
138. Finite Element Seminar Series, Tianyuan Mathematical Center in Northeast China, 2020/6 (Online talk)
139. Computational Mathematics Seminar, School of Mathematical Sciences, East Normal University of China, 2020/7 (Online talk)
140. Colloquium, School of Applied Mathematics and Sciences, Beijing University of Technology, China, 2020/7 (Online talk)
141. Zhufeng Colloquium, School of Mathematics, Shandong University, China, July, 2020/10 (Online talk)

Contributed

1. Postdoc Seminar, IMA, University of Minnesota, 2002/11
2. Brown Bag Seminar, IMA, University of Minnesota, 2003/10
3. Postdoc Seminar, IMA, University of Minnesota, 2004/2
4. Scientific Computing Seminar, Department of Mathematics, University of South Carolina, 2004/5
5. Scientific Computing Seminar, Department of Mathematics, University of South Carolina, 2005/3
6. ACM Seminar, Department of Mathematics, University of South Carolina, 2009/10
7. ACM Seminar, Department of Mathematics, University of South Carolina, 2011/9

Conference/Workshop Presentations

Invited Talks

1. *"Finite volume methods on spherical Voronoi meshes"*, Emerging Methodologies and Applications in Numerical PDEs, Tallahassee, FL, March 2004
2. *"Cortical surface flattening using discrete conformal mapping with minimal metric distortion"*, The Second IEEE International Symposium on Biomedical Imaging, Arlington, VA, April 2004
3. *"Finite volume methods on the sphere and spherical centroidal Voronoi meshes"*, SIAM Annual Meeting 2005, New Orleans, LA, July 2005
4. *"Adaptive finite element method based on conforming centroidal Voronoi Delaunay triangulations"*, SIAM Southeastern Atlantic Section Annual Meeting 2006, Auburn, AL, March 2006
5. *"Adaptive finite element method based on conforming centroidal Voronoi Delaunay triangulations"*, International Conference on Recent Advance in Scientific Computation, Beijing, China, June 2006
6. *"Adaptive methods for PDEs and conforming centroidal Voronoi Delaunay triangulations"*, SIAM Conference on Computational Science and Engineering 2007, Costa Mesa, CA, February 2007
7. *"Nondegeneracy and weak global convergence of Lloyd algorithm"*, SIAM Annual Meeting 2008, San Diego, CA, July 2008

8. *"A multi-resolution method for climate system modeling"*, SIAM Conference on Computational Science and Engineering 2009, Miami, FL, March, 2009
9. *"Adaptive finite volume methods for steady convection-diffusion equations with mesh optimization"*, SIAM Southeastern Atlantic Section Annual Meeting 2009, Columbia, SC, April 2009
10. *"Coupled models and parallel simulations for three-dimensional full-Stokes ice sheet modeling"*, Spring AMS Southeastern Section Meeting 2010, Lexington, KY, March 2010
11. *"Coupled models and parallel simulations for three-dimensional full-Stokes ice sheet modeling"*, SIAM Southeastern Atlantic Section Annual Meeting 2010, Raleigh, NC, March 2010
12. *"Coupled models and parallel simulations for three-dimensional full-Stokes ice sheet modeling"*, International Workshop on Scientific Computing and Nonlinear Partial Differential Equations, Jiuzhaigou, Sichuan Province, China, June 2010
13. *"A parallel high-order accurate finite element full-Stokes ice-sheet model with validation using benchmark experiments"*, SIAM Southeastern Atlantic Section Annual Meeting 2011, Charlotte, NC, March 2011
14. *"A parallel high-order accurate finite element full-Stokes ice-sheet model and benchmark experiments"*, 2011 Spring AMS Western Section Meeting Las Vegas, NV, April 2011
15. *"A parallel high-order accurate finite element nonlinear Stokes ice-sheet model and benchmark experiments"*, International Conference on Applied Mathematics and Interdisciplinary Research, Tianjin, China, June 2011
16. *"A multichannel edge-weighted centroidal Voronoi tessellation algorithm for 3D superalloy image segmentation"*, International Congress on Industrial and Applied Mathematics 2011, Vancouver, Canada, July 2011
17. *"A parallel high-order accurate finite element full-Stokes ice-sheet model with validation using benchmark experiments"*, International Congress on Industrial and Applied Mathematics 2011, Vancouver, Canada, July 2011
18. *"Image segmentation by centroidal Voronoi tessellation based models"*, International Congress on Industrial and Applied Mathematics 2011, Vancouver, Canada, July 2011
19. *"Covolume-upwind finite volume approximations for linear elliptic partial differential equations"*, SIAM Southeastern Atlantic Section Annual Meeting 2012, Huntsville, AL, March 2012
20. *"Covolume-upwind finite volume approximations for linear elliptic partial differential equations"*, International Conference on Scientific Computing and Applications 2012, Las Vegas, NV, April 2012
21. *"A posteriori error analysis of finite element method for linear nonlocal diffusion and peridynamic models"*, The Second International Conference on Scientific Computing, Nanjing, China, May 2012
22. *"A posteriori error analysis of finite element method for linear nonlocal diffusion and peridynamic models"*, International Workshop on Computational and Applied Mathematics, Beijing, China, June 2012
23. *"Image segmentation methods based on centroidal Voronoi tessellation and its variations"*, International Workshop on Numerical Methods for Partial Differential Equations, Jinan, China, June 2012
24. *"A posteriori error analysis of finite element method for linear nonlocal diffusion and peridynamic models"*, Nonlocal Continuum Models for Diffusion, Mechanics, and Other Applications, SAMSI, Durham, NC, June 2012
25. *"A posteriori error analysis of finite element method for linear nonlocal diffusion and peridynamic models"*, International Conference on Mathematical Modeling, Analysis and Computation, Xiamen, China, July 2012

26. *"A posteriori error analysis of finite element method for linear nonlocal diffusion and peridynamic models"*, AMS and MAA Joint Mathematics Meetings 2013, San Diego, CA, January 2013

27. *"An efficient and accurate parallel computational model for 3d thermo-mechanical Stokes ice-sheet flow simulation"*, CESM-LIWG Winter Meeting 2013, Boulder, CO, February 2013

28. *"Image segmentation methods based on centroidal voronoi tessellation and its variants"*, SIAM Conference on Computational Science and Engineering 2013, Boston, MA, February-March 2013

29. *"A parallel computational model for three-dimensional thermo-mechanical Stokes flow simulations of glaciers and ice sheets"*, SIAM Southeastern Atlantic Section Annual Meeting 2013, Knoxville-ORNL, TN, March 2013

30. *"A parallel computational model for three-dimensional thermo-mechanical Stokes flow simulations of glaciers and ice sheets"*, International Conference on Mathematical Modeling and Computation, Wuhan, China, May 2013

31. *"Centroidal Voronoi tessellation based methods for image segmentation"*, International Workshop on Computational and Applied Mathematics, Yellow Mountain, China, May 2013

32. *"Fast explicit integration factor methods for semilinear parabolic equations"*, The Second International Conference on Engineering and Computational Mathematics, Hong Kong, December 2013

33. *"Fast explicit integration factor methods for semilinear parabolic equations"* , Spring AMS Southeastern Section Meeting 2014, Knoxville, TN, March 2014

34. *"A parallel computational model for three-dimensional thermo-mechanical Stokes flow simulations of glaciers and ice sheets"* , SIAM Southeastern Atlantic Section Annual Meeting 2014, Melbourne, FL, March 2014

35. *"Fast explicit integration factor methods for semilinear parabolic equations"*, The 2014 International Conference on Modeling of Complex Biological Systems, Tianjin, China, May 2014

36. *"A parallel computational model for three-dimensional thermo-mechanical Stokes flow simulations of glaciers and ice sheets"*, International Workshop on the Frontiers of Computational Geodynamics, Beijing, China, July 2014

37. *"Fast explicit integration factor methods for semilinear parabolic equations"*, Fall AMS Southeastern Section Meeting 2014, Greensboro, NC, November, 2014

38. *"A finite element three-dimensional stokes ice sheet dynamics model with enhanced local mass conservation"*, SIAM Conference on Computational Science and Engineering 2015, Salt Lake City, Utah, March 2015

39. *"Fast and accurate algorithms for simulating coarsening dynamics of Cahn-Hilliard equations"*, SIAM Southeastern Atlantic Section Annual Meeting, Birmingham, AL, March 2015

40. *"Fast and accurate algorithms for simulating coarsening dynamics of Cahn-Hilliard equations"*, SIAM Central States Section Annual Meeting 2015, Rolla, MO, April 2015

41. *"Fast and accurate algorithms for simulating coarsening dynamics of Cahn-Hilliard equations"*, International Conference on Numerical Partial Differential Equations and Their Applications, Wuhan, China, May 2015

42. *"Efficient and stable exponential time difference Runge-Kutta methods for phase field elastic bending energy models"*, International Congress on Industrial and Applied Mathematics 2015, Beijing, China, August 2015

43. *"Nonlocal convection-diffusion problems and finite element approximations"*, International Congress on Industrial and Applied Mathematics 2015, Beijing, China, August 2015

44. *"A finite element three-dimensional Stokes ice sheet dynamics model with enhanced local mass conservation"*, Workshop on Computational Mathematics and Scientific Computing, Jeju Island, South Korea, August 2015

45. *"Centroidal Voronoi tessellation based methods for image segmentation"*, Workshop on Advances in Scientific Computing and Applied Mathematics, Las Vegas, NV, October 2015

46. *"Efficient and stable exponential time difference Runge-Kutta methods for phase field elastic bending energy models"*, SIAM Southeastern Atlantic Section Annual Meeting 2016, Athens, GA, March 2016

47. *"Efficient and stable exponential time difference Runge-Kutta methods for phase field elastic bending energy models"*, International Conference on Applied Mathematics, Hong Kong, May-June 2016

48. *"Efficient and stable exponential time difference Runge-Kutta methods for phase field elastic bending energy models"*, Workshop on Computational Mathematics and Application Advances, Jiujiang, Jiangxi, China, July 2016

49. *"Efficient and stable exponential time difference Runge-Kutta methods for phase field elastic bending energy models"*, SIAM Central States Section Annual Meeting 2016, Little Rock, AK, September-October 2016

50. *"Exponential time differencing schemes for the epitaxial growth model without slope selection"*, SIAM Central States Section Annual Meeting 2016, Little Rock, AK, September-October 2016

51. *"Exponential time differencing gauge method for incompressible viscous flows"*, IMACS 2016 World Congress, Xiamen, China, December 2016

52. *"A parallel computational model for 3D thermo-mechanical Stokes flow simulations of ice sheets"*, Workshop on Current Tends in Numerical PDEs and Applications, Xi'an, China, December 2016

53. *"Exponential time differencing gauge method for incompressible viscous flows"*, AMS-MAA Joint meeting 2017, Atlanta, January 2017

54. *"Scalable compact localized exponential time differencing method for simulating coarsening dynamics of Cahn-Hilliard equations"*, SIAM Conference on Computational Science and Engineering 2017, Atlanta, GA, March 2017

55. *"A conservative nonlocal convection-diffusion model and asymptotically compatible finite difference discretization"*, SIAM Southeastern Atlantic Section Annual Meeting 2017, Tallahassee, FL, March 2017

56. *"Stabilized compact exponential time differencing methods for gradient flow problems and scalable implementation"*, Workshop on Advances of Scientific Computing, Nanchang University, China, May 2017

57. *"A conservative nonlocal convection-diffusion model and asymptotically compatible finite difference discretization"*, Third International Conference on Engineering and Computational Mathematics, Hong Kong, May-June 2017

58. *"Stabilized compact exponential time differencing methods for gradient flow problems and scalable implementation"*, Workshop on Phase Field Method and Applications, Beijing International Center of Mathematical Research, China, July 2017

59. *"Fast spherical centroidal Voronoi mesh generation: a Lloyd-preconditioned LBFGS method in parallel"*, The 16th International Workshop on Multi-scale (Un)-structured Mesh Numerical Modeling, Stanford University, CA, August 2017

60. *"Fast spherical centroidal Voronoi mesh generation: a Lloyd-preconditioned LBFGS method in parallel"*, Workshop on Complex Systems and High-Performance Computing, Xuchang, China, September 2017

61. *"Overlapping localized exponential time differencing methods for diffusion problems"*, Workshop on Computational Mathematics, Hong Kong Polytechnic University, Hong Kong, December 2017
62. *"Overlapping localized exponential time differencing methods for diffusion problems"*, International Conference on Recent Advances in Computational and Applied Mathematics, Wuhan, China, December 2017
63. *"Conservative explicit local time-stepping schemes for the shallow water equations"*, Workshop on Modeling and Algorithms for Ocean Simulations, Beijing, China, May 2018
64. *"Conservative explicit local time-stepping schemes for the shallow water equations"*, Workshop on High Accuracy Schemes and Conservative Schemes for PDEs, Qingdao, China, June 2018
65. *"Conservative explicit local time-stepping schemes for the shallow water equations"*, The Twelfth AIMS Conference on Dynamical Systems, Differential Equations and Applications, Taipei, Taiwan, July 2018
66. *"Maximum principle preserving exponential time differencing schemes for the nonlocal Allen-Cahn equation"*, The Twelfth AIMS Conference on Dynamical Systems, Differential Equations and Applications, Taipei, Taiwan, July 2018
67. *"Localized exponential time differencing methods: algorithms and analysis"*, Workshop on High Performance and Scientific Computing, Qingdao, China, October 2018 (**Keynote speaker**)
68. *"Localized exponential time differencing methods: algorithms and analysis"*, Workshop on Mathematical and Computational Methods for Quantum Systems, Montreal, Canada, December 2018
69. *"Conservative explicit local time-stepping schemes for the shallow water equations"*, International Youth Forum on Computational Mathematics, Xiangtan, China, December 2018
70. *"Maximum principle preserving exponential time differencing schemes for the nonlocal Allen-Cahn equation"*, SIAM Conference on Computational Science and Engineering 2019, Spokane, WA, February-March 2019
71. *"Localized exponential time differencing methods: algorithms and analysis"*, Workshop on Numerical Computing and Applications, Hunan Normal University, Changsha, China, May 2019
72. *"Localized exponential time differencing methods: algorithms and analysis"*, International Conference on Advanced Numerical Methods for Scientific Computation, Southern University of Science and Technology, Shenzhen, China, June 2019
73. *"Conservative explicit local time-stepping schemes for the shallow water equations"*, International Workshop on Numerical Methods, Scientific Computing and Applications, Yantai University, China, July 2019
74. *"Conservative explicit local time-stepping schemes for the shallow water equations"*, The Twelfth China National Conference on Computational Mathematics, Harbin, China, July-August 2019 (**Plenary speaker**)
75. *"High order explicit local time-stepping methods for hyperbolic conservation laws"*, SIAM Southeastern Atlantic Section Annual Meeting 2019, Knoxville, TN, September 2019
76. *"Multi-video temporal synchronization by matching pose features of shared moving subjects"*, The ICCV Workshop on Moving Cameras, Seoul, Korea, October 2019
77. *"Localized exponential time differencing methods for shallow water equations: algorithms and numerical investigations"*, AMS-MAA Joint Meeting 2020, Denver, CO, January 2020
78. *"Conservative explicit local time-stepping schemes for the shallow water equations"*, Scientific Computing Around Louisiana (SCALA) 2020, Baton Rouge, LA, February 2020 (**Keynote speaker**)

79. *"A conservative nonlocal convection-diffusion model and asymptotically compatible finite difference discretization"*, Workshop on Recent Progress in Nonlocal Modeling, Analysis and Computation, June 2020 (Visual conference)
80. *"Maximum bound principles for semilinear parabolic equations and exponential time differencing schemes"*, OUC Workshop on Computational Mathematics, July 2020 (Visual conference)
81. *"Parallel exponential time differencing methods for ocean dynamics"*, DOE ESMD-E3SM PI Meeting, October 2020 (Visual conference)
82. *"Localized exponential time differencing methods: algorithms and analysis"*, Workshop on Recent progress in Phase Field Related Problems, November 2020 (Visual conference)
83. *"Stabilized integrating factor Runge-Kutta method and unconditional preservation of maximum bound principle"*, SIAM Conference on Computational Science and Engineering 2019, March 2021 (Visual conference)

<u>Posters</u>

1. *"Cortical surface flattening using discrete conformal mapping with minimal metric distortion"*, Human Brain Mapping 2003 (HBM'03), New York, June 2003
2. *"Quantitative evaluation of three cortical surface flattening Methods"*, Human Brain Mapping 2004 (HBM'04), Budapest, Hungary, June 2004
3. *"ParcelMan and Pigment: interactive tools for parcellation of cortical grey matter"*, Human Brain Mapping 2004 (HBM'04), Budapest, Hungary, June 2004
4. *"New grid and discretization technologies for ocean and ice simulations"*, Climate Change Prediction Program Meeting 2007, Indianapolis, IN, September 2007
5. *"Conserving energy and potential vorticity on unstructured Voronoi diagrams"*, Newton Institute Scoping Meeting on Multi-scale Modelling of the Atmosphere and Ocean, University of Reading, UK, March 2009
6. *"A parallel solver for three dimensional full-Stokes ice sheet modeling"*, Climate Change Prediction Program Meeting 2009, Bethesda, MD, April 2009
7. *"Conserving energy and potential vorticity on unstructured Voronoi diagrams"*, Climate Change Prediction Program Meeting 2009, Bethesda, MD, April 2009
8. *"3D open-surface shape correspondence for statistical shape modeling: Identifying topologically consistent landmarks*, The Twelfth IEEE International Conference on Computer Vision, Kyoto, Japan, September-October 2009
9. *"Over-relaxation Lloyd method for computing centroidal Voronoi tessellations"*, Cha-Cha Days Workshop 2009, Orlando, FL, November 2009
10. *"Greenland ice-sheet simulations via parallel three-dimensional full-Stokes modeling"*, DOE Climate Change Modeling Program Science Team Meeting 2010, Gaithersburg, MD, March-April 2010
11. *"Nonuniform centroidal Voronoi grids for climate simulations"*, DOE Climate Change Modeling Program Science Team Meeting 2010, Gaithersburg, MD, March-April 2010
12. *"Model for prediction across scale"*, DOE Climate Change Modeling Program Science Team Meeting 2010, Gaithersburg, MD, March-April 2010
13. *"A multichannel edge-weighted centroidal Voronoi tessellation algorithm for 3D super-alloy image segmentation"*, The Twenty-Fourth IEEE Conference on Computer Vision and Pattern Recognition, Colorado Springs, CO, June 2011
14. *"Grain segmentation in 3D superalloy images using multichannel EWCVT under human annotation constraints"*, The Twelfth European Conference on Computer Vision, Firenze, Italy, October 2012

15. *"Multiscale superpixels and supervoxels based on hierarchical edge-Weighted centroidal Voronoi tessellation"*, The IEEE Winter Conference on Applications of Computer Vision, Waikoloa Beach, HI, January 2015
16. *"Fast CVT grid generation for ocean modeling: a Lloyd-preconditioned L-BFGS method in parallel"*, DOE ACME All-Hands Meeting, Potomac, MD, June 2017
17. *"Exponential time differencing for large time stepping and localized approach for parallel implementation"*, DOE ACME All-Hands Meeting, Potomac, MD, June 2017
18. *"Conservative explicit local time-stepping schemes for the shallow water equations"*, DOE Modeling PI's Meeting, Potomac, MD, November 2018
19. *"Localized exponential time differencing methods based on domain decomposition"*, DOE Modeling PI's Meeting, Potomac, MD, November 2018
20. *"Semantic stereo matching with pyramid cost volumes"*, The Seventeenth International Conference on Computer Vision, Seoul, Korea, October-November 2019.
21. *"P-MVSNet: Learning patch-wise matching confidence aggregation for multi-view stereo"*, The Seventeenth International Conference on Computer Vision, Seoul, Korea, October-November 2019.
22. *"Spatial correspondence with generative adversarial network: learning depth from monocular videos"*, The Seventeenth International Conference on Computer Vision, Seoul, Korea, October-November 2019.

Organized Meetings

Conferences/Workshops

1. Member of organizing committee, SIAM Southeastern Atlantic Section Annual Meeting 2008, Orlando, FL, March 2008
2. Co-Chair of organizing committee, SIAM Southeastern Atlantic Section Annual Meeting 2009, Columbia, SC, April 2009
3. Co-Chair of organizing committee, International Workshop on Computational and Applied Mathematics, Beijing, China, June 2012
4. Member of organizing committee, International Workshop on Complex Systems and High Performance Computing, Xuchang, China, September 2017
5. Co-Chair of organizing committee, Ninth Annual Graduate Student Mini-conference in Computational Mathematics, Columbia, SC, February 2018
6. Co-Chair of organizing committee, International Conference on Mathematical Modeling and Numerical Methods, Qingdao, China, May-June 2019

Minisymposia

1. Organizer, Minisymposium – *"Algorithms and Applications of Tessellations of Spaces and Surface"* (1 session), SIAM Southeastern Atlantic Section Annual Meeting 2006, Auburn, AL, March 2006
2. Co-Organizer, Minisymposium – *"Centroidal Voronoi Tessellations: Theory, Algorithms and Applications"* (2 sessions), SIAM Conference on Computational Science & Engineering 2007, Costa Mesa, CA, February 2007
3. Organizer, Minisymposium – *"Numerical Methods for Partial Differential Equations"* (2 sessions), SIAM Southeastern Atlantic Section Annual Meeting 2008, Orlando, FL, March 2008
4. Co-Organizer, Minisymposium – *"Recent Advances in Algorithms and Applications of Centroidal Voronoi Tessellation and Optimal Quantization"* (2 sessions), SIAM Annual Meeting 2008, San Diego, CA, July 2008

5. Co-Organizer, Minisymposium – *"Recent Advances in Studies and Applications of Centroidal Voronoi Tessellations"* (2 sessions), International Congress on Industrial and Applied Mathematics 2011, Vancouver, Canada, July 2011.
6. Co-Organizer, Minisymposium – *"Recent Advances in Numerical PDEs and Computational Biology"* (2 sessions), SIAM Southeastern Atlantic Section Annual Meeting 2012, Huntsville, AL, March 2012
7. Co-Organizer, Minisymposium – *"Recent Advances in Modeling of Complex Systems: Analysis and Computation"* (3 sessions), SIAM Southeastern Atlantic Section Annual Meeting 2014, Melbourne, FL, March 2014
8. Co-Organizer, Minisymposium – *"Modeling and simulations of Complex Biological Systems"* (4 sessions), International Congress on Industrial and Applied Mathematics 2015, Beijing, China, August 2015
9. Co-Organizer, Minisymposium – *"Recent Advances on Modeling, Analysis and Computation of Nonlocal Problems "* (3 sessions), SIAM Southeastern Atlantic Section Annual Meeting 2017, Tallahassee, March 2017
10. Co-Organizer, Workshop – *"Numerical Methods for Phase Field Models"*, The Third International Conference on Engineering and Computational Mathematics, Hong Kong, May-June 2017
11. Organizer, Minisymposium – *"Numerical Partial Differential Equations and Applications"* (1 session), SIAM Southeastern Atlantic Section Annual Meeting 2018, Chapel Hill, NC, March 2018

NSF Committee

- NSF-DMS Panelist for Comput. Math. Program, March 2008
- NSF-DMS Panelist for Comput. Math. Program, March 2010
- NSF-DMS Panelist for Career Program, November 2012
- NSF-DMS Panelist for Comput. Math. Program, March 2013
- NSF-DMS Panelist for Comput. Math. Program, March 2014
- NSF-DMS Panelist for Comput. Math. Program, March 2016

Referee for Proposals

- National Science Foundation
- Singapore National Research Foundation
- Ministry of Education Research Grant, Singapore
- Hong Kong Research Grants Council
- National Research Foundation, United Arab Emirates
- KAUST Competitive Research Grants

Referee for Professional Journals

Advances in Applied Mathematics and Mechanics
Advances in Engineering Software
Applied Mathematics & Computation
Applied Numerical Mathematics
Communications in Computational Physics
Computational Materials Science
Computer-Aided Design
Computers & Fluids
Computers & Mathematics with Applications
Computer Methods in Applied Mechanics and Engineering
Computer Physics Communications

CVPR 2020, 2021
Discrete & Continuous Dynamical Systems B
East Asia Journal of Applied Mathematics
Engineering with Computers
IEEE Access
IEEE Internet of Things Journal
IEEE Network Magazine
IEEE Transactions on Industrial Informatics
IEEE Transactions on Information Technology in BioMedicine
IEEE Transactions on Medical Imaging
IEEE Transactions on Pattern Analysis and Machine Intelligence
IEEE Transactions on Visualization and Computer Graphics
International Journal for Numerical Methods in Engineering
International Journal for Uncertainty Quantification
International Journal of Computer Mathematics
International Journal of Numerical Analysis and Modeling
ISRN Applied Mathematics
Journal of Atmospheric and Oceanic Technology
Journal of Computational and Applied Mathematics
Journal of Computational Mathematics
Journal of Computational Physics
Journal of Integral Equations and Applications
Journal of Mathematical Analysis and Applications
Journal of Scientific Computing
Ocean Dynamics
Ocean Engineering
Pattern Recognition
Pattern Recognition Letters
Mathematics of Computation
Mathematical Methods in the Applied Sciences
Mechanics Research Communications
Multiscale Modeling and Simulations
NeurIPS 2020
Numerical Linear Algebra with Applications
Numerical Mathematics: Theory, Methods and Applications
Numerische Mathematik
Science in China
SIAM Journal on Applied Dynamical Systems
SIAM Journal on Applied Mathematics
SIAM Journal on Numerical Analysis
SIAM Journal on Scientific Computing
SIGGRAPH 2015
WACV 2021

Research Supervision

Graduate Students

- Doctoral Students
  - ☐ Li Tian (USC/Math), Ph.D., 2009/8.
    Dissertation: *"Error Estimates for Finite Element/Volume Approximations of Dissipative Partial Differential Equations on Surfaces"*
  - ☐ Xiao Xiao (USC/Math), Ph.D., 2012/5.

Dissertation: *"Covolume-upwind Finite Volume Approximations for Linear Elliptic Partial Differential Equations"*

- ☐ Yu Cao (USC/CSE), Ph.D. (co-supervisor), 2013/8.
  Dissertation: *"3D Grain Segmentation in Superalloy Images Using Multichannel Edge-Weighted Centroidal Tessellation Based Methods"*
- ☐ Youjie Zhou (USC/CSE), Ph.D. (co-supervisor), 2015/8.
  Dissertation: *"Propagated Image Segmentation Using Edge-Weighted Centroidal Voronoi Tessellation Based Methods"*
- ☐ Chenfei Zhang (USC/Math), Ph.D., 2019/8.
  Dissertation: *"Unconditionally Energy Stable Linear Schemes for a Two-Phase Diffuse Interface Model with Peng-Robinson Equation of State"*
- ☐ Shuai Yuan (USC/Math), Ph.D.(co-supervisor), 2020/8.
  Dissertation: *"An Ensemble-based Projection Method and its Numerical Investigation"*
- ☐ Yuankai Teng (USC/Math), Ph.D., current.
- ☐ Zhenyao Wu (USC/CSE), Ph.D.(co-supervisor), current.
- ☐ Xinyi Wu (USC/CSE), Ph.D.(co-supervisor), current.

- Masters Students
  - ☐ Wensong Wu (USC/Math), M.S., 2007/8.
    Thesis: *"Adaptive Finite Volume Methods for Convection-Diffusion Equations with Mesh Optimization"*
  - ☐ Xiao Xiao (USC/Math), M.S., 2010/5.
    Thesis: *"Over-relaxation Lloyd Method for Computing Centroidal Voronoi Tessellations"*
  - ☐ Jing Liu (USC/Math), M.S., 2010/8.
    Thesis: *"Construction of Centroidal Voronoi Tessellations Using a Conjugate Gradient Method Based on Trust Regions"*
  - ☐ Rosalia Tatano (USC/Math), M.S., 2013/5.
    Thesis: *"Study on Covolume-Upwind Finite Volume Approximations for Linear Parabolic Partial Differential Equations"*
  - ☐ Meshack Kiplagat (USC/Math), M.S., 2013/12.
    Thesis: *"The Compact Implicit Integration Factor Scheme for the Solution of Allen-Cahn Equation"*

Postdocs

- Huai Zhang, Postdoc, 2008/9-2010/3.
- Xiaoran Shi, Postdoc (at BCSRC), 2012/7-2013/7.
- Hao Tian, Postdoc (at BCSRC), 2013/7-2016/1.
- Tong Zhang, Postdoc, 2013/8-2015/8.
- Jianfang Lu, Postdoc (at BCSRC), 2016/7-2018/8.
- Thi-Thao-Phuong Hoang, Postdoc, 2017/1-2018/8.
- Chunbao Zhou, Postdoc, 2017/1-2017/8.
- Huadong Gao, Postdoc, 2018/3-2018/12.
- Xiao Li, Postdoc, 2018/8-2019/8.
- Xucheng Meng, Postdoc, 2018/9-2019/8.

- Rihui Lan, Postdoc, 2020/2-Present.

Visiting Scholars

- Yang Li, Visiting Student, 2010/8-2011/2.
- Weidong Zhao, Visiting Scholar, 2013/9-2013/11.
- Wei Leng, Visiting Scholar, 2013/9-2013/12.
- Liyong Zhu, Visiting Scholar, 2014/3-2015/3.
- Huai Zhang, Visiting Scholar, 2015/8-2015/9.
- Jian Zhang, Visiting Scholar, 2015/10-2015/11.
- Hongwei Li, Visiting Scholar, 2016/9-2017/9.
- Xiaoping Zhang, Visiting Scholar, 2018/6-2018/8.
- Jianguo Huang, Visiting Scholar, 2018/8-2018/9.
- Zhaocan Ma, Visiting Student, 2019/2-2019/5.
- Kun Jiang, Visiting Student, 2019/9-2020/9.
- Jingwei Li, Visiting Student, 2019/10-2020/10.
- Anthony Gruber, Visiting Scholar, 2020/1-Present.

Teaching at USC

Number	Course Title	Crs.	Terms
MATH 141	Calculus I	4	S06, S10, F11
MATH 142	Calculus II	4	S12, S14, F19, F20
MATH 241	Vector Calculus	3	F04, F12, F14, F18
MATH 242	Elementary Differential Equations	3	S09, S13, S15, S16, S19, S20
MATH 344	Applied Linear Algebra	3	F18, S19
SCHC 499A	Senior Thesis/Project	3	S10, S11
MATH 514	Financial Mathematics I	3	F13, F19
MATH 515	Financial Mathematics II	3	S14
MATH 520	Ordinary Differential Equations	3	F07, F15
MATH 526	Numerical Linear Algebra	3	S05, S06, F06, F08, F09, S12, S16
MATH 527	Numerical Analysis	3	S10
MATH 550	Vector Analysis	3	S20, S21
MATH 706	Numerical Linear Algebra	3	S08
MATH 708	Foundations of Comput. Math. I	3	F11, F12
MATH 709	Foundations of Comput. Math. II	3	S13, S20
MATH 723	Differential Equations I	3	F14, F16
MATH 724	Differential Equations II	3	S15, S17
MATH 726	Numerical Analysis I	3	F06
MATH 727	Numerical Analysis II	3	S07
MATH 728J	Scientific Computing I	3	F05
MATH 798	Directed Readings & Research	1-3	Su07, Su12, Su15
MATH 799	Thesis Research	1-3	S07, Su07, S10, Su10, S13, F13, Su16
MATH 899	Dissertation Research	1-3	F07, S08, Su08, F08, S09, S10, Su10, F10, S11, Su11, F11, S12, Su17, F17, S18, Su18, F18, S19, Su19, F19

Service

Department

- Chair of Committee of the Tenured Faculty (2009/3 – 2010/4)
- Post-Tenure Review Committee (2008/8 – 2009/3, 2011/8 – 2013/4, 2014/8 – 2016/8)
- Undergraduate Advisory Council (2004/8 – 2006/8)
- Graduate Advisory Council (2006/8 – 2007/8, 2009/8 – 2010/8, 2011/8 – Present)
- Faculty Advisory Council (2012/8 – 2014/8, 2018/8 – Present)
- Hiring Committee (2005/8 – 2006/8, 2007/8 – 2009/8)
- BioMath Hiring Committee (2012/8 – 2013/8)
- Applied and Computational Mathematics Committee (2004/8 – 2010/8, 2011/8 – Present)
- Calculus II Coordinator (2020/8 – Present)
- Graduate Placement Committee (2011/8 – 2013/8)
- Peer Review of Teaching Committee (2016/9 – Present)
- Grant Mentoring Committee (2009/8 – 2010/8)
- Computer Committee (2004/8 – 2010/8)
- Faculty Advisor of SIAM Student Chapter at USC (2006/8 – 2008/8)
- Organizer of Applied and Computational Mathematics Seminar (2009/8 – 2010/8)
- Editor for the Industrial Mathematics Institute preprint series (2005/4 – 2010/8)
- Editor for the Mathematics Undergraduate Research preprint series (2005/8 – 1010/8)
- Undergraduate Advisor (2005/8 – 2006/8)
- Coach for the Mathematical Contest in Modeling (2006/2 – 2008/8)
- Organizer of the SIAM Student Chapter Seminar (2006/8 – 2008/8)
- Doctoral Dissertation Committees of 16 Ph.D. students since 2004
    - ☐ 2005: Yabin Ding (USC/Math), Teresa Jin (USC/Math)
    - ☐ 2006: Kening Wang (USC/Math), Shuang Li (USC/Math)
    - ☐ 2007: Luke Owens (USC/Math)
    - ☐ 2009: Sarthok Sircar (USC/Math), Mark Walters (USC/Math)
    - ☐ 2011: Mingrui Yang (USC/Math), Pahal Dalal (USC/CSE), Jian Shi (USC/CSE)
    - ☐ 2012: Zhiqi Zhang (USC/CSE)
    - ☐ 2015: Jia Zhao (USC/Math), Che Wang (USC/Math), Qiujin Peng (HK Polytech/Math)
    - ☐ 2016: Xiaochuan Fan (USC/CSE), Yuewei Lin (USC/CSE)
    - ☐ 2019: Shuang Liu (USC/Math)
    - ☐ 2020: Xiangcheng Zheng (USC/Math)
- Judge for the Student Research Competition of SIAM Student Chapter at USC (2005/3)

College and University

- Member of IMI Executive Committee (2012/9 – Present)
- Member of USC General Education Work Team "Life-Long Learning" led by Professor Cynthia Colbert (2007/1 – 2007/8)
- Member of University Committee of Tenure and Promotion (2014/8 – 2014/12)

Community

- Board memeber of the SC East Point Academy (2015/8-2017/6)

- Judge for 2005 USC Region II Science and Engineering Fair (Math./Comput. Sci. Section)
- Judge for the 19th (2005/4), 20th (2006/1), 21st (2006/12), 22nd (2007/12), 23rd (2008/12), 24th (2010/1), 26th (2012/1), 28th (2014/1), 29th (2015/1), 30th (2016/1), 31st (2017/1), 33rd (2019/1), 34th (2020/1) USC High School Math Contest

<u>Professional</u>

- External research evaluator for 11 Promotion and/or Tenure cases at peer institutions
  - ☐ Department of Mathematics and Statistics (Kening Wang, Associate Professor&Tenure), University of North Florida, 2011
  - ☐ Department of Mathematical Sciences (Maria Emelianenko, Associate Professor&Tenure), George Mason University, 2011
  - ☐ Department of Mathematics and Statistics (Yanzhi Zhang, Associate Professor&Tenure), Missouri University of Science and Technology, 2014
  - ☐ Department of Mathematical Sciences (Leo Rebholz, Associate Professor&Tenure), Clemson University, 2014
  - ☐ Department of Mathematics (Yanping Ma, Associate Professor&Tenure), Loyola Marymount University, 2016
  - ☐ Department of Mathematics (Steven Wise, Full Professor), University of Tennessee at Knoxville, 2016
  - ☐ Department of Mathematics (Yanxiang Zhao, Associate Professor&Tenure), George Washington University, 2018
  - ☐ Department of Mathematics (Ying Wang, Associate Professor&Tenure), University of Oklahoma, 2018
  - ☐ Department of Mathematical Sciences (Qingshan Chen, Associate Professor&Tenure), Clemson University, 2018
  - ☐ Department of Mathematics (Yanlai Chen, Full Professor), University of Massachusetts at Dartmouth, 2019 & 2020
  - ☐ Department of Mathematics and Statistics (Hans Werner van Wyk, Associate Professor&Tenure), Auburn University, 2020
- Served in six NSF panels
- Reviewed proposals for NSF, DOE and some international agencies
- Editorial service for several journals
- Reviewed papers for over 50 journals and conference proceedings
- Organized/Co-Organized 6 international and national meetings and 11 minisymposia in conferences
- Member of Hiring Committee of Institute of Mathematical Sciences at Renmin University of China (2013 – 2015)